

ReLIFE Forecasting Service / NEIs from Indoor Thermal Conditions

Wiki documentation text for the linear heat and cold DALY calculation

PROJECT	
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Project acronym:	HARMONISE
Project name:	Harm Assessment On Indoor Short-Term Exposures

Revision and history chart:

Version	Date	Editors	Comment
1	10/07/2026	GiM	Shared with DaA. Draft for Wiki integration.
2			

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1 Inputs, parameters, and equations

This section documents the calculation logic used by the code. The method converts daily indoor air temperature into an expected health burden expressed in disability-adjusted life years, or DALYs. In this context, the health burden is interpreted as a non-energy impact (or health co-benefit) related to indoor thermal conditions.

The current implementation is intentionally simple. It uses only general-population, linear heat and cold harm-intensity factors for a Mediterranean/Italian context. It does not include elderly-specific parameters, log-linear formulations, uncertainty propagation, or sensitivity analysis.

Important implementation rule: *thresholds must be selected as predefined heat-cold pairs. The user should not be allowed to combine a heat threshold from one scenario with a cold threshold from another scenario.*

Scope	General-population, linear heat and cold harm-intensity calculation only. Default coefficients are for a Mediterranean/Italian context.
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1.1 Required inputs

Input	Unit / format	Meaning	Implementation note
Daily indoor air temperature	degrees C	Daily mean indoor air temperature for the assessed building, zone, or dwelling.	Each valid row should represent one day. If the model produces hourly or sub-hourly values, aggregate them to daily mean values before using this version of the calculation.
Threshold-pair scenario	text / option	Defines the heat and cold thresholds used to classify the daily temperature.	Select one complete pair only. Do not mix thresholds across pairs.
Population	persons	Number of people represented by the assessed profile.	Used only to scale the per-person result to a population-level DALY result.
Exposure days for selected period	days	Number of days used for the annual or selected-period total.	For example, 360 or 365 days. This does not need to be the same as the number of pasted rows.
Heat linear HI	DALY / degrees C / 100,000 persons / day	Harm-intensity coefficient for heat exposure.	Fixed parameter in the current tool.
Cold linear HI	DALY / degrees C / 100,000 persons / day	Harm-intensity coefficient for cold exposure.	Fixed parameter in the current tool.

Table 1: Required inputs for the linear heat and cold DALY calculation.

1.2 Threshold pairs

The tool uses paired thresholds. This is central to the interpretation. For a given scenario, the heat threshold defines where heat harm starts, and the cold threshold defines where cold harm starts. Daily temperatures between the two thresholds are interpreted as a neutral zone and produce zero thermal harm in this simplified linear implementation.

Scenario option	Heat threshold (degrees C)	Cold threshold (degrees C)	Interpretation	Recommended use
ITALY_MMT_24_4	24.4	24.4	Single minimum-mortality-temperature benchmark for Italy. Temperatures above 24.4 degrees C are treated as heat exposure, and temperatures below 24.4 degrees C are treated as cold exposure.	Health-risk attribution using the Italy MMT benchmark.
COMFORT_PAIR_26_20	26.0	20.0	Indoor comfort pair. Heat harm starts above 26.0 degrees C. Cold harm starts below 20.0 degrees C. Values between 20.0 and 26.0 degrees C return zero harm.	Comfort-oriented comparison of indoor thermal performance.
CVD_PAIR_30_21	30.0	21.0	Cardiovascular-health-oriented pair. Heat harm starts above 30.0 degrees C. Cold harm starts below 21.0 degrees C. Values between 21.0 and 30.0 degrees C return zero harm.	Health-oriented screening where CVD-related thermal thresholds are preferred.

Table 2: Heat and cold threshold pairs used in the linear DALY calculation.

1.3 Harm-intensity coefficients used in this version

The harm-intensity coefficient, or HI, expresses the expected DALY burden caused by one degree Celsius of daily temperature deviation for one day in a population of 100,000 persons.

The default HI values in this version are for a Mediterranean/Italian context. They use Mediterranean risk estimates and Italy-level health burden data. They are not universal values. For other countries or climate regions, the HI values and thresholds should be replaced with other values.

Because the coefficient is expressed per 100,000 persons, population scaling must always divide by 100,000 before multiplying by the selected population.

Thermal domain	Linear HI	Unit	Population group	Health endpoint	Default context
Heat	0.7152769	DALY / degrees C / 100,000 persons / day	All ages / general population	Non-accidental mortality	Mediterranean risk + Italy burden data
Cold	0.3481684	DALY / degrees C / 100,000 persons / day	All ages / general population	Cardiovascular-disease mortality	Mediterranean risk + Italy burden data

Table 3: Linear harm-intensity coefficients used in the current Mediterranean/Italian-context version of the tool.

1.4 Equations

For each valid daily temperature row i , define:

- T_i = daily mean indoor air temperature for day i (degrees C)
- $heat_threshold_C$ = selected heat threshold (degrees C)
- $cold_threshold_C$ = selected cold threshold (degrees C)
- HI_heat = 0.7152769 DALY / degrees C / 100,000 persons / day
- HI_cold = 0.3481684 DALY / degrees C / 100,000 persons / day
- P = selected population (persons)
- CF = 100,000 persons

1.4.1 Step 1. Calculate heat and cold deviations

$heat_deviation_i = \text{MAX}(0, T_i - heat_threshold_C)$

$cold_deviation_i = \text{MAX}(0, cold_threshold_C - T_i)$

If T_i is between the selected cold and heat thresholds, both deviations are zero. If the selected pair is ITALY_MMT_24_4, the same value is used for heat and cold; values above 24.4 degrees C produce heat harm, and values below 24.4 degrees C produce cold harm.

1.4.2 Step 2. Calculate daily harm rate per 100,000 persons

$heat_harm_100k_i = heat_deviation_i * HI_heat$

$cold_harm_100k_i = cold_deviation_i * HI_cold$

$total_harm_100k_i = heat_harm_100k_i + cold_harm_100k_i$

1.4.3 Step 3. Convert to per-person daily harm

$heat_harm_person_i = heat_harm_100k_i / CF$

$cold_harm_person_i = cold_harm_100k_i / CF$

$total_harm_person_i = total_harm_100k_i / CF$

1.4.4 Step 4. Scale to the selected population

$heat_harm_population_i = heat_harm_person_i * P$

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cold_harm_population_i = cold_harm_person_i * P
total_harm_population_i = total_harm_person_i * P

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1.4.5 Step 5. Summarise across the profile or selected period

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average_daily_heat_harm_population = MEAN(heat_harm_population_i)
pasted_profile_heat_harm_population = SUM(heat_harm_population_i)
annual_or_period_heat_harm_population = average_daily_heat_harm_population * exposure_days_for_period

```

The same aggregation applies to cold harm and total harm. The pasted-profile total should be used when the input profile itself represents the assessment period. The annual or selected-period total should be used when the pasted profile is treated as a representative profile and the user wants to scale it to a defined number of exposure days.

1.5 Calculating health co-benefits or avoided harm

The tool can estimate health co-benefits when two indoor temperature profiles are compared. The baseline profile represents indoor temperatures before the intervention, retrofit, control strategy, or forecasted change. The intervention profile represents indoor temperatures after that action. The comparison must use the same threshold pair, population, and exposure period for both profiles.

- avoided_heat_DALY = baseline_heat_DALY - intervention_heat_DALY
- avoided_cold_DALY = baseline_cold_DALY - intervention_cold_DALY
- avoided_total_DALY = baseline_total_DALY - intervention_total_DALY

A positive avoided DALY value means that the intervention reduces expected health burden. This is interpreted as a health co-benefit. A negative value means that the intervention increases expected health burden (added harm / disbenefit) under the selected threshold pair and assumptions.

2 How to read and interpret the outputs

This section is intended for the user-facing Wiki page. It explains the outputs shown by the tool and how users should interpret them.

2.1 General interpretation of thermal harm (as DALYs)

A DALY is a measure of health loss. One DALY corresponds to one year of healthy life lost in a population. In this tool, DALYs are not observed deaths or diagnosed cases. They are expected statistical health burdens estimated from epidemiological coefficients and indoor temperature exposure. Daily per-person DALY values are usually small. This is expected. Thermal health impacts accumulate across days, buildings, and populations. Therefore, the population-scaled and annual or selected-period totals are often easier to interpret than a single daily per-person value. For communication, DALYs can also be converted to healthy-life days:

- $\text{healthy_life_days_lost} = \text{DALY} * 365$

2.2 Output table guide

The Linear_tool sheet reports expected heat, cold, and total DALY harm for one indoor temperature profile.

Output shown by the tool	Unit	Meaning	How to interpret it
Average daily heat harm rate	DALY / 100,000 persons / day	Mean heat-related harm rate across valid daily temperature rows.	Use this to compare heat risk across scenarios independently of the selected population size.
Average daily cold harm rate	DALY / 100,000 persons / day	Mean cold-related harm rate across valid daily temperature rows.	Use this to compare cold risk across scenarios independently of the selected population size.
Average daily total harm rate	DALY / 100,000 persons / day	Heat plus cold harm rate.	This gives the average daily total thermal health burden at the normalized 100,000-person scale.
Average daily heat harm per person	DALY / person / day	Average heat harm rate divided by 100,000.	This is the daily expected heat burden for one person under the represented temperature profile.
Average daily cold harm per person	DALY / person / day	Average cold harm rate divided by 100,000.	This is the daily expected cold burden for one person under the represented temperature profile.
Average daily total harm per person	DALY / person / day	Heat plus cold per-person daily burden.	Use this only for individual-level scaling. It remains an expected statistical burden, not an individual prediction.
Average daily heat harm for population	DALY / day	Average per-person heat harm multiplied by the selected population.	This is the expected daily heat burden for the population represented by the tool input.
Average daily cold harm for population	DALY / day	Average per-person cold harm multiplied by the selected population.	This is the expected daily cold burden for the population represented by the tool input.

Average daily total harm for population	DALY / day	Heat plus cold daily burden for the selected population.	Use this to compare the average daily health burden of alternative scenarios.
Pasted-profile heat harm for population	DALY for pasted profile	Sum of daily heat burden across all valid pasted rows.	Use this when the pasted temperature profile itself is the full assessment period.
Pasted-profile cold harm for population	DALY for pasted profile	Sum of daily cold burden across all valid pasted rows.	Use this when the pasted temperature profile itself is the full assessment period.
Pasted-profile total harm for population	DALY for pasted profile	Heat plus cold burden across the pasted profile.	This is the total expected burden represented directly by the input profile.
Annual or selected-period heat harm for population	DALY / selected period	Average daily population heat harm multiplied by the selected exposure days.	Use this when the input profile is a representative period and must be scaled to a year or another defined duration.
Annual or selected-period cold harm for population	DALY / selected period	Average daily population cold harm multiplied by the selected exposure days.	Use this when the input profile is a representative period and must be scaled to a year or another defined duration.
Annual or selected-period total harm for population	DALY / selected period	Heat plus cold burden for the selected period.	This is the main value for annual or period-level co-benefit comparison.

Table 4: Single-profile outputs reported by the Linear_tool sheet and their interpretation.

2.3 Comparison output table guide

The Comparison_tool sheet compares two indoor temperature profiles: a baseline profile and an intervention profile. The baseline profile represents indoor temperatures before the intervention, retrofit, control action, or forecasted change. The intervention profile represents indoor temperatures after that action.

The tool reports baseline harm, intervention harm, and avoided DALYs. Avoided DALYs are calculated as: $\text{avoided_DALYs} = \text{baseline_DALYs} - \text{intervention_DALYs}$

A positive avoided value means lower expected harm after the intervention. This is interpreted as avoided harm or a health co-benefit. A negative avoided value means higher expected harm after the intervention. This is interpreted as added harm or a disbenefit.

Output column	Meaning	How to interpret it
Baseline	DALY harm calculated from the indoor temperature profile before the intervention/action.	This is the reference health burden.
Intervention	DALY harm calculated from the indoor temperature profile after the intervention/action.	This is the health burden after the assessed change.
Avoided	Baseline minus intervention.	Positive values indicate avoided harm. Negative values indicate added harm.
Check	Plain-language interpretation of the avoided value.	Shows whether the result is a health co-benefit, added harm/disbenefit, or no change.

Table 5: Comparison outputs reported by the Comparison_tool sheet and interpretation of avoided DALYs.

2.4 Practical reading rules for users

- Read heat and cold separately first. Heat and cold may respond differently to the same building intervention.
- Use the paired-threshold scenario label when reporting results. A result without its threshold pair is incomplete.
- Do not add the pasted-profile total and the annual or selected-period total. They are alternative aggregations of the same daily harm calculation.
- Use normalized outputs per 100,000 persons for comparison across cases with different population sizes.
- Use population-scaled outputs when the question concerns the expected burden for a specific building, district, or user-defined population.
- Use annual or selected-period totals when reporting non-energy impacts or health co-benefits over a defined period.
- When comparing a baseline and an intervention, avoided DALYs are calculated as baseline DALYs minus intervention DALYs.
- For comparison mode, read heat, cold, and total avoided DALYs separately. An intervention can reduce cold harm while increasing heat harm, or the opposite. The total avoided DALY value shows the net effect. It shows the overall shift in the balance of trade-offs.

2.5 Example wording for the Wiki

The following text can be used directly in the Wiki page:

The heat and cold DALY outputs represent expected health burden from daily indoor air temperature deviations beyond selected thresholds. Heat harm is calculated only when daily indoor temperature is above the selected optimal temperature threshold. Cold harm is calculated only when daily indoor temperature is below the selected optimal temperature threshold. If the temperature is between the selected thresholds, the daily thermal harm is zero in this implementation. Results are first calculated as DALYs per 100,000 persons per day, then converted to per-person and population-scaled DALYs using the selected population size. Period totals are obtained by multiplying the average daily population burden by the selected number of exposure days.

2.6 Example interpretation of one output set

If the tool reports an average daily heat harm rate of 3.117 DALY / 100,000 persons / day and zero cold harm, this means that the temperature profile produces heat exposure above the selected heat threshold but no cold exposure below the selected cold threshold. Dividing by 100,000 gives 0.00003117 DALY / person / day. For a population of one person and a 360-day exposure period, this corresponds to about 0.0112 DALY, or about 4.1 healthy-life days lost during the selected period. For larger populations, the same per-person burden is multiplied by the population size.

2.7 Example interpretation of comparison outputs

The Comparison_tool compares a baseline temperature profile with an intervention temperature profile. The baseline profile represents indoor temperatures before the intervention, retrofit, control action, or forecasted change. The intervention profile represents indoor temperatures after that action. The main comparison value is the **avoided DALY**:

$$- \text{avoided_DALY} = \text{baseline_DALY} - \text{intervention_DALY}$$

The total avoided DALY is the main value for judging the overall health co-benefit or disbenefit. The heat avoided DALY and cold avoided DALY explain how each thermal domain contributes to the total result.

2.7.1 Positive avoided DALYs: health co-benefit

If the baseline annual/period total harm is 0.00391 DALY and the intervention annual/period total harm is 0.00257 DALY, then:

$$- \text{avoided_total_DALY} = 0.00391 - 0.00257 = 0.00134 \text{ DALY}$$

The positive value means that the intervention reduces the expected total thermal health burden. This is interpreted as **avoided harm** or a **health co-benefit**.

Converted to healthy-life days:

$$- 0.00134 \text{ DALY} \times 365 = 0.49 \text{ healthy-life days}$$

This means that, for the selected population and period, the intervention avoids about 0.49 healthy-life days lost compared with the baseline.

The heat and cold avoided DALYs should still be read separately. They show whether the benefit comes from reduced heat harm, reduced cold harm, or both.

2.7.2 Zero or near-zero avoided DALYs: no net change

If the baseline annual/period total harm is 0.00300 DALY and the intervention annual/period total harm is also 0.00300 DALY, then:

$$- \text{avoided_total_DALY} = 0.00300 - 0.00300 = 0.00000 \text{ DALY}$$

This means that the intervention does not change the expected total thermal health burden under the selected threshold pair and assumptions.

A zero total can occur in two ways. First, heat and cold harm may both remain unchanged. Second, one domain may improve while the other worsens by a similar amount. For example, the intervention may reduce cold harm but increase heat harm by the same DALY amount. In that case, the net health balance is zero, but the thermal profile has changed.

The total avoided DALY shows the net health effect. Heat and cold avoided DALYs explain the components of that net result.

2.7.3 Negative avoided DALYs: added harm or disbenefit

If the baseline annual/period total harm is 0.00257 DALY and the intervention annual/period total harm is 0.00391 DALY, then:

– $\text{avoided_total_DALY} = 0.00257 - 0.00391 = -0.00134 \text{ DALY}$

The negative value means that the intervention increases the expected total thermal health burden. This is interpreted as **added harm** or a **health disbenefit**.

Converted to healthy-life days:

– $-0.00134 \text{ DALY} \times 365 = -0.49 \text{ healthy-life days}$

This means that, for the selected population and period, the intervention adds about 0.49 healthy-life days lost compared with the baseline.

This can happen when the intervention improves one thermal domain but worsens the other by a larger amount. For example, an intervention may reduce cold harm but increase heat harm more strongly. In that case, the total avoided DALY becomes negative, meaning that the intervention is not beneficial overall under the selected threshold pair and assumptions

2.8 Limitations and wording to keep visible

- This is a screening and scenario-comparison tool, not an individual clinical risk model.
- The current version uses general-population linear harm-intensity coefficients only.
- The default coefficients are for a Mediterranean/Italian context. They should not be treated as universal values.
- For use in other countries or climate regions, the threshold pairs and harm-intensity coefficients should be replaced with locally appropriate values.
- The current version does not include elderly-specific coefficients, log-linear risk functions, probabilistic uncertainty, or sensitivity analysis.
- The calculation assumes daily mean indoor air temperature as the exposure metric.
- The heat and cold coefficients are based on selected health endpoints. They should not be interpreted as the complete burden of all possible temperature-related health outcomes.
- The results depend strongly on the selected threshold pair. The threshold pair must always be reported with the output.
- The method applies population-level epidemiological parameters to indoor temperature profiles as a first-order approximation. This is useful for scenario comparison, but it should not be interpreted as observed local disease incidence.

3 References

- Morantes Quintana, G. et al. Estimating indoor heat-related health impacts in social housing using DALYs: a preliminary framework. IEQ / ASHRAE conference paper, 2025. DOI: 10.63044/eq25mo23
- Morantes Quintana, G. et al. A preliminary framework for cold-related health impacts in social housing using DALYs. Indoor Air conference paper, 2026. DOI: *tbd*

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The project HARMONISE – Harm Assessment on Indoor Short-Term Exposures, awarded under the EU Horizon Europe 2024 Marie Skłodowska-Curie Postdoctoral Fellowship call, is funded by the Autonomous Province of Bolzano/Bozen through the Seal of Excellence funding scheme (Decree No. 4400/2025).